

Zhenyuan Dong

Ph.D. Student in Electrical and Computer Engineering, Purdue University

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EDUCATION

Purdue University

Ph.D. in Electrical and Computer Engineering

Research focus: Computer Vision, Representation Learning

Advisor: [Qiang Qiu](#)

West Lafayette, IN

Aug. 2025 – May 2030 (Expected)

New York University

M.S. in Computer Engineering

Research focus: Efficient Machine Learning

New York, NY

Sep. 2023 – May 2025

Shanghai University

B.Eng. in Computer Science

Research focus: Machine Learning

Shanghai, China

Sep. 2019 – Jun. 2023

RESEARCH INTERESTS

His research interests include machine learning and computer vision, with a focus on generative models, self-supervised learning, and representation learning. In particular, he is interested in representation learning in both feature space and parameter space for generative models and world modeling, especially on post-training and test-time learning.

PUBLICATIONS

* Equal contribution

MANUSCRIPTS UNDER REVIEW

2026

Coefficient-Calibrated Memory Retrieval via Sparse Coding for Long Video Generation

Zhenyuan Dong, Wei Chen, Jingxi Yu, Qiang Qiu

Under review.

PEER-REVIEWED PUBLICATIONS

NeurIPS 2026

Approaching Effective Merging in Model Embedding Space

Wei Chen, Zhenyuan Dong, Qiang Qiu

Advances in Neural Information Processing Systems (NeurIPS 2026).

AAAI 2026

Self-Supervised Representation Learning with Joint Embedding Predictive Architecture for Automotive LiDAR Object Detection

Haoran Zhu, Zhenyuan Dong, Kristi Topollai, Beiyao Sha, Anna Choromanska

Proceedings of the AAAI Conference on Artificial Intelligence (AAAI 2026).

WACV 2025

DiTAS: Quantizing Diffusion Transformers via Enhanced Activation Smoothing

Zhenyuan Dong, Sai Qian Zhang

IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2025).

Findings of
EMNLP 2025

Mitigating Sequential Dependencies: A Survey of Algorithms and Systems for Generation-Refinement Frameworks in Autoregressive Models

Yunhai Hu*, Zining Liu*, Zhenyuan Dong*, Tianfan Peng*, Bradley McDanel, Sai Qian Zhang

Findings of the Conference on Empirical Methods in Natural Language Processing (EMNLP 2025 Findings).

RESEARCH EXPERIENCE

Coefficient-Calibrated Memory Atom Retrieval via Sparse Coding for Long Video Generation

Research Assistant, Purdue University

Advisor: [Qiang Qiu](#)

- Developed CARE-MEM, a training-free memory interface for long-video generation, improving Dynamic Degree by 3.48 points on average over LongLive-RAG while preserving long-range consistency across three autoregressive backbones.

May 2026 – Sep. 2026

West Lafayette, IN

- Proposed *constrained sparse coding optimization for memory selection* to retrieve complementary memory atoms under a fixed memory budget.
- Introduced *coefficient-calibrated priors for attention control*, leveraging sparse coefficients to modulate how strongly each selected memory atom participates in attention.

Sep. 2025 – Apr. 2026

Model Embedding Space for Effective Model Merging

Research Assistant, Purdue University

West Lafayette, IN

Advisor: [Qiang Qiu](#)

- Developed a model embedding space derived from pretrained weights that constrains downstream adaptation to a shared low-dimensional coordinate system, improving mergeability while preserving pretrained structure.
- Proposed SVD-based aligned subspace merging to reduce interference among independently adapted models; validated on ViT-B/32 and MM-MergeBench with stronger merged-model performance and fewer trainable parameters than LoRA.

Jan. 2024 – Jan. 2025

Quantization on Diffusion Transformers for Efficient Large Generative Model Inference

Research Assistant, New York University

New York, NY

Advisor: [Sai Qian Zhang](#)

- Proposed DiTAS, a data-free post-training quantization method for efficient Diffusion Transformer inference, achieving state-of-the-art image-generation results under low-bit quantization.
- Designed temporal-aggregated smoothing, layer-wise grid search, and LoRA-based alternating optimization to reduce quantization error without additional fine-tuning.

Feb. 2024 – Jan. 2025

Self-Supervised World Models for Autonomous Driving with Joint Embedding Predictive Architecture

Research Assistant, New York University

New York, NY

Advisor: [Anna Choromanska](#)

- Introduced AD-L-JEPA, a self-supervised joint-embedding predictive architecture that learns spatiotemporal representations from multi-frame LiDAR data for autonomous-driving world modeling.
- Achieved state-of-the-art label-efficient performance, outperforming supervised world-model baselines with more accurate predictions and stronger generalization across driving scenarios.

Feb. 2022 – Jan. 2023

Energy-Efficient Brain-Like Artificial Intelligence via Spiking Neural Networks

Research Assistant, Shanghai University

Shanghai, China

Advisor: [Shi Gu](#)

- Studied ANN-to-SNN conversion using progressive tandem learning and parameter calibration for energy-efficient brain-inspired representation learning.
- Achieved 90.87% accuracy on the CIFAR-10 test set with the hybrid SNN model, outperforming conventional SNN networks trained with surrogate-gradient descent.

HONORS & AWARDS

Myron M. Rosenthal Award for Best MS Academic Achievement, New York University

May 2025

Graduate Scholarship, Tandon School of Engineering, New York University

Sep. 2023 – May 2025

Best Thesis, School of Computer Engineering and Science, Shanghai University

Jul. 2023

Outstanding Graduate of Shanghai University – Top 5%

Jul. 2023

National Silver Medal, China Collegiate Computing Contest – Programming Tournament – Top 1%

May 2022

22nd CCF Computer Software Awards – Top 10%

Apr. 2021

Outstanding Student of Shanghai University – Top 5%

Dec. 2020

PROFESSIONAL SERVICES

Conference Reviewer: CVPR 2026; NeurIPS 2026; AAAI 2026, 2027.

TECHNICAL SKILLS

Programming & ML: Python, PyTorch, CUDA, C/C++, MATLAB, Pandas, NumPy, OpenCV, Git, Java.

Research Tools: $\text{\LaTeX}$, Linux. **Languages:** Chinese (native); English (working proficiency).